MILITARY SPECIFICATION

RESISTORS, VARIABLE, NONWIRE-WOUND, STYLE RVC6

This specification is approved for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers the detail requirements for style RVC6 resistors.

2. APPLICABLE DOCUMENTS

2.1 The following document, of the issue in effect on date of invitation for bids or request for proposal, forms a part of this specification to the extent specified herein.

SPECIFICATION

MILITARY

MIL-R-23285 - Resistors, Variable, Nonwire-Wound, General Specification for.

(Copies of specifications, standards, drawings, and publications required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

3. REQUIREMENTS

3.1 Requirements. Requirements shall be in accordance with MIL-R-23285, and as specified herein.

3.2 Design and construction. Resistors shall be of the design, construction, and physical dimensions shown on figure.

3.2.1 Mounting and locking nuts. The mounting nut shall be 5/64 inch thick and shall measure 5/16 inch across the hexagonal flats. For locking bushings, the locking nut shall be 5/32 inch thick and shall measure 5/16 inch across the hexagonal flats. Thread size shall be 1/4-32NEF-2B.

3.2.2 Internal-tooth lockwasher. An internal-tooth lockwasher shall be supplied and when mounted, shall have a minimum thickness of approximately 0.045 inch.

3.2.3 Retainer rings. If retainer rings are used, they shall be not thicker than 0.032 inch.

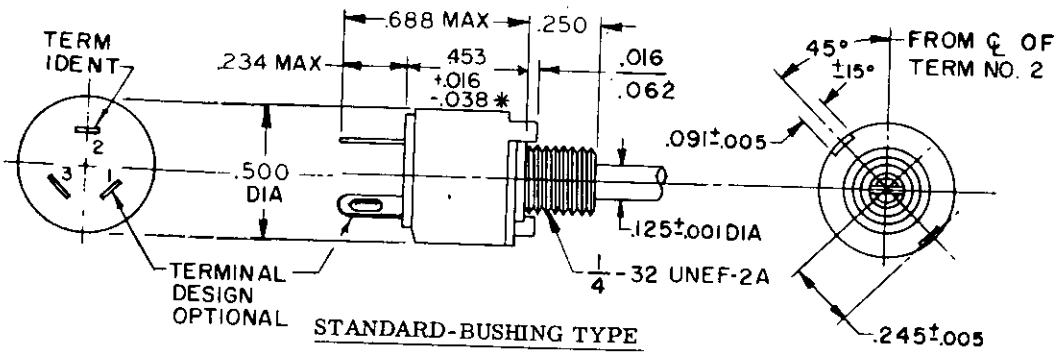
3.2.4 Operating shaft length. The operating shaft length shall be in accordance with table I.

TABLE I. Length of operating shaft.

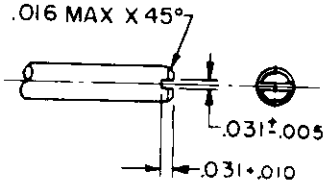
Symbol	Nominal length of shaft measured from mounting surface of resistors. 1/	
	Bushings N and S Inches ±1/32	Bushings L and T Inches ±1/32
A	---	5/8
D	7/8	---
L	3/8	---

1/ For resistors with panel seals. The mounting surface shall be considered the metal face of the panel seal when the seal is firmly seated against the resistor body.

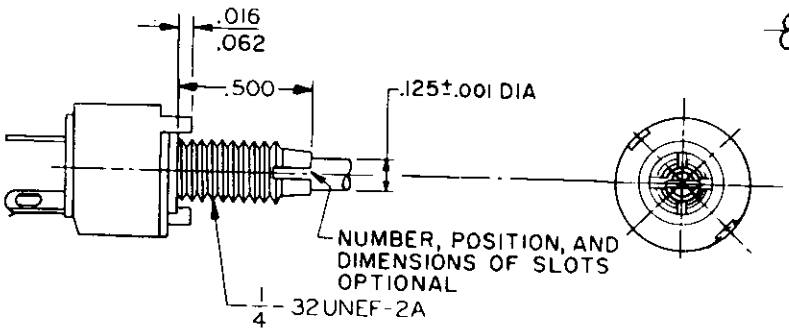
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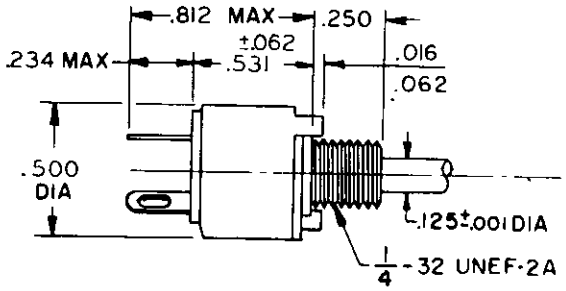
STANDARD-BUSHING TYPE



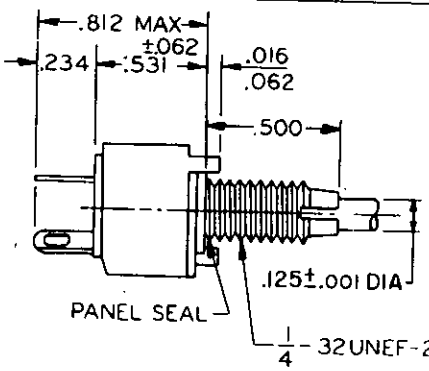
SLOTTED SHAFT



LOCKING-BUSHING TYPE



SHAFT-AND PANEL-SEALED-STANDARD BUSHING TYPE



SHAFT AND PANEL-SEALED-LOCKING BUSHING TYPE

INCHES	MM
.001	.03
.005	.13
.010	.25
.016	.41
.031	.79
.038	.97
.062	1.57
.091	2.31
.125	3.18
.234	5.94
.245	6.22
.250	6.35
.453	11.51
.500	12.70
.531	13.49
.688	17.48
.812	20.62

NOTES:
1. Dimensions are in inches.
2. Metric equivalents (to the nearest .01 mm) are given for general information only and are based upon 1 inch = 25.4 mm.
3. Unless otherwise specified, tolerance is .XXX ±.016 inch (.41 mm).

STYLE RVC6 Resistors.

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3.3 Total mechanical rotation. The total mechanical rotation without switch shall be within the limits of 292° and 298°.

3.4 Nominal total resistance and rated continuous working voltage. The nominal total resistance and rated continuous working voltage (RCWV) shall be as specified in table II.

TABLE II. Nominal total resistance and rated continuous working voltage.

Nominal total resistance Ohms	RCWV[1] (volts) Taper-A	Taper-C	Nominal total resistance Megohms	RCWV[1] (volts) Taper-A	Taper-C
100	7	---	.10	224	160
250	11	---	.25	350	200
500	16	---	.50	350	200
1,000	22	16	1.0	350	200
2,500	35	25	2.0	350	---
5,000	50	36	2.5	350	---
10,000	71	50			
25,000	112	80			
50,000	158	112			

[1] Rated continuous working voltage at 125° C.

3.5 Maximum weight. The maximum weight shall be 25 grams.

3.6 Minimum resistance. The maximum value of minimum resistance shall be 15 ohms for total resistance values of 100 to 500 ohms inclusive.

3.7 Power rating. The power rating shall be 1/2 watt for taper-A resistors and 1/4 watt for taper-C resistors.

3.8 Torque.

3.8.1 Operating. The torque required to effect rotation shall be not less than 0.50 ounce-inch nor more than 6 ounce-inches.

3.8.2 Stop. The torque applied through the operating shaft to the stops shall be 3 pound-inches.

3.8.3 Locking bushing. For the locking-bushing type, the locking nut shall be tightened with a torque of 8 pound-inches. After the locking nut is tightened, the contact arm shall not move when a torque of 20 ounce-inches is applied to the shaft.

4. QUALITY ASSURANCE PROVISIONS

4.1 Sampling and inspection. Sampling and inspection shall be in accordance with MIL-R-23285, and as specified herein.

4.2 Dielectric withstanding voltage. In the dielectric withstanding voltage test, the applied potential shall be 350 volts root mean square (rms) at reduced barometric pressure.

5. PREPARATION FOR DELIVERY

5.1 Preparation for delivery shall be in accordance with MIL-R-23285.

6. NOTES

6.1 Notes. The notes specified in MIL-R-23285 are applicable to this specification.

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6.2 Changes from previous issue. The margins of this specification are marked with an asterisk to indicate where changes (additions, modifications, corrections, deletions) from the previous issue were made. This was done as a convenience only and the Government assumes no liability whatsoever for any inaccuracies in these notations. Bidders and contractors are cautioned to evaluate the requirements of this document based on the entire content irrespective of the marginal notations and relationship to the last previous issue.

Custodians:
Army - EL
Navy - EC
Air Force - 11

Review activities:
Army - MI, MU
Navy - EC
Air Force - 11, 17, 85
DSA - ES

User activities:
Army - ME, AT, AV
Navy -
Air Force -

Preparing activity:
Army - EL

Agent:
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